

WICK, United Kingdom

WMO: 030750

Lat: 58.4539N

Lon: 3.0900W

Elev: 126

StdP: 14.63

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.5	29.1	22.2	16.8	30.7	24.6	18.8	32.9	40.2	40.2	35.9	40.1	7.0	260	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	64.8	59.1	62.7	57.8	61.1	56.7	60.4	63.5	59.0	61.5	57.8	60.1	13.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.9	74.7	61.8	57.6	71.2	60.1	56.6	68.6	59.0	26.8	63.9	25.8	61.6	25.0	60.2	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.4	29.5	26.3	DB	21.7	71.0	4.1	2.3	18.7	72.7	16.3	74.0	13.9	75.3	10.9	77.0	(4)
(5)			WB	20.9	63.9	4.0	1.6	18.0	65.0	15.7	66.0	13.4	66.8	10.5	68.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	39.3	39.2	41.0	44.0	47.2	51.7	55.2	55.4	53.0	48.5	43.5	39.9	(6)	
	DBStd	7.34	4.29	4.67	4.61	4.37	4.09	3.77	3.48	3.25	3.91	4.49	4.94	5.40	(7)	
	HDD50	1841	332	303	280	185	105	24	2	2	16	80	198	314	(8)	
	HDD65	6738	797	722	744	629	552	399	305	297	360	511	644	778	(9)	
	CDD50	578	0	0	1	5	18	76	163	170	106	35	3	1	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	Wind	WSAvg	12.9	15.1	15.0	14.7	13.3	11.7	11.2	10.1	10.4	12.0	13.7	14.3	14.0	(14)
	Precipitation	PrecAvg	31.0	2.9	2.3	2.5	1.9	1.9	2.3	2.8	3.0	3.0	3.6	3.1	(15)	
(16)		PrecMax	39.6	5.5	5.3	4.6	3.8	3.2	4.2	3.9	5.9	6.3	6.9	5.4	(16)	
(17)		PrecMin	22.1	0.9	0.6	0.6	0.5	0.4	0.5	0.9	0.7	0.9	0.8	1.1	(17)	
(18)		PrecStd	4.3	1.1	1.0	0.9	0.8	0.7	0.8	0.8	1.3	1.1	1.3	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.5	53.8	55.4	58.8	61.5	65.0	69.0	67.8	65.9	60.0	55.3	53.4	(19)	
		MCWB	48.9	50.3	50.2	52.5	55.2	57.7	62.0	61.2	59.5	56.1	52.1	50.9	(20)	
	2%	DB	48.9	50.6	51.7	54.5	58.0	61.5	64.8	64.1	62.1	57.3	53.2	50.9	(21)	
		MCWB	46.8	48.0	47.9	49.6	52.8	55.9	59.6	58.8	57.0	53.9	50.7	48.8	(22)	
	5%	DB	47.1	48.2	49.5	52.1	55.4	59.4	62.7	62.2	60.0	55.8	51.9	49.2	(23)	
		MCWB	45.2	45.8	46.2	47.9	51.1	54.7	58.3	57.6	55.9	53.0	49.9	47.2	(24)	
	10%	DB	45.5	46.0	47.6	50.1	53.4	57.5	60.9	60.6	58.4	54.6	50.6	47.6	(25)	
		MCWB	43.8	43.7	44.7	46.7	49.8	53.6	57.0	56.7	55.0	52.1	48.7	45.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	51.0	51.1	53.6	56.6	59.3	63.3	62.1	60.2	57.2	53.1	51.2	(27)	
		MCDB	50.8	53.2	54.5	57.4	60.3	63.2	67.1	66.1	64.2	59.1	54.2	52.8	(28)	
	2%	WB	47.2	48.5	48.5	50.3	53.7	57.2	60.9	60.1	58.3	54.9	51.6	49.2	(29)	
		MCDB	48.6	50.2	51.1	53.6	56.8	60.4	63.9	63.0	60.9	56.4	52.7	50.7	(30)	
	5%	WB	45.6	46.1	46.7	48.6	51.9	55.7	59.2	58.6	56.8	53.8	50.3	47.5	(31)	
		MCDB	46.9	48.1	48.9	51.3	54.6	58.4	61.6	60.9	58.8	55.2	51.6	48.9	(32)	
	10%	WB	44.1	44.1	45.1	47.3	50.4	54.3	57.8	57.6	55.8	52.7	48.9	45.9	(33)	
		MCDB	45.4	45.7	47.1	49.6	52.8	56.8	60.1	59.6	57.7	54.3	50.4	47.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.4	9.5	9.8	10.5	9.7	9.6	9.7	9.5	8.8	8.0	8.0	(35)	
		MCDBR	8.7	10.2	11.6	12.0	12.3	12.0	12.0	11.3	10.9	9.3	8.4	9.2	(36)	
		MCWBR	8.2	8.7	9.0	8.0	8.2	7.9	7.8	7.4	7.6	7.8	7.8	8.8	(37)	
	5% WB	MCDBR	8.4	9.6	11.0	10.7	11.0	11.2	10.8	9.8	9.2	8.3	8.0	8.9	(38)	
		MCWBR	8.2	8.7	9.1	7.7	7.9	8.0	7.8	7.2	7.1	7.8	7.9	8.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.317	0.340	0.360	0.379	0.372	0.369	0.376	0.365	0.353	0.340	0.319	0.300	(40)	
	taud		2.194	2.303	2.330	2.315	2.389	2.429	2.421	2.458	2.484	2.408	2.242	2.131	(41)	
	Ebn at Noon		156	216	247	262	273	275	270	265	250	214	154	120	(42)	
	Edh at Noon		16	23	29	34	34	33	33	30	25	20	15	12	(43)	
(44) All-Sky Solar Radiation	RadAvg		116	308	650	1124	1537	1514	1434	1149	766	396	154	74	(44)	
	RadStd		11	33	49	91	93	109	95	72	63	32	13	8	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page